

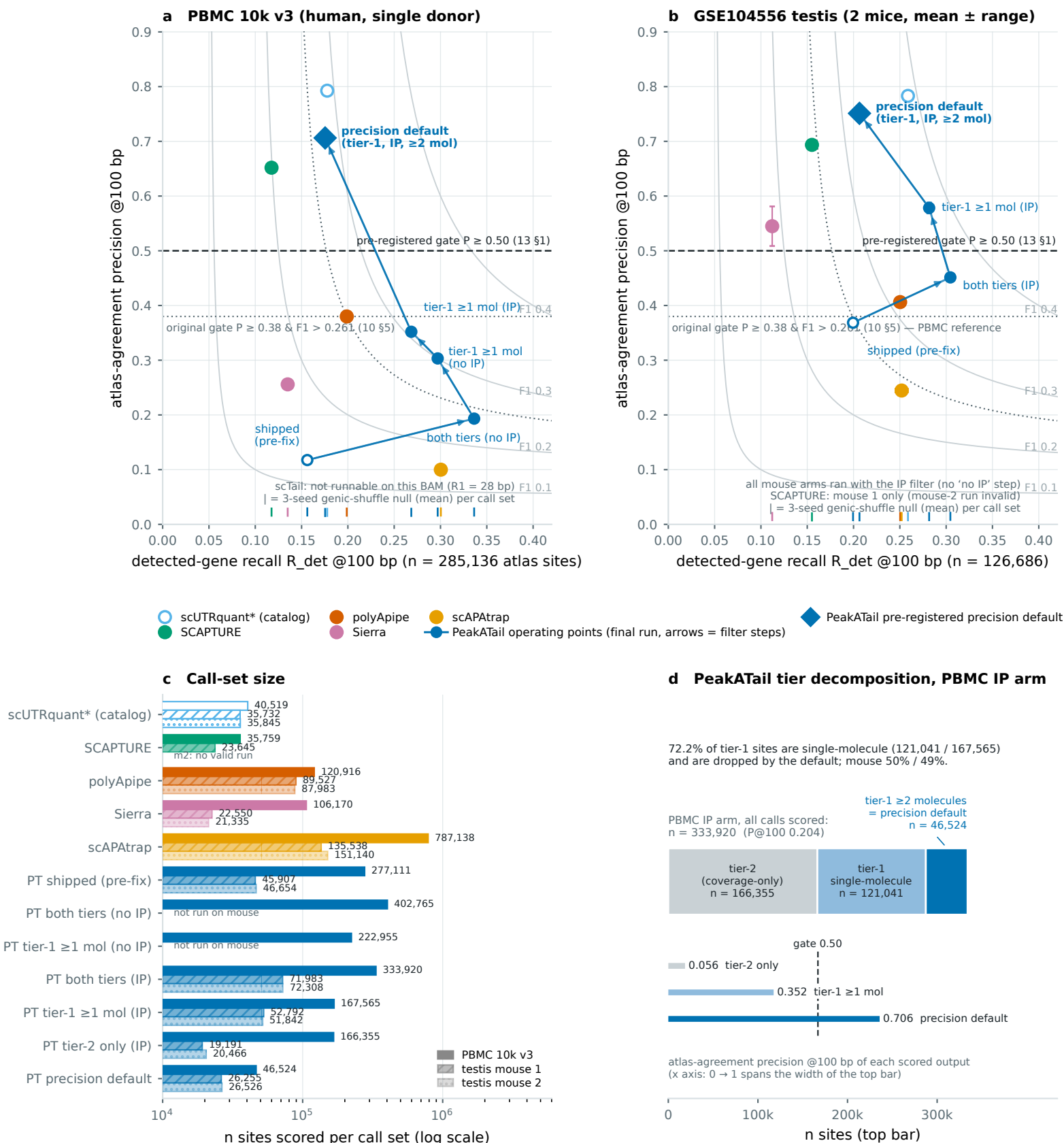


Fig 2 | Trustworthy PAS detection: the final Stage-2 run (PeakATail code 9dfdefb, #96 + #97 merged) against the competitor panel. Definitions: atlas-agreement precision = fraction of calls within 100 bp (point, strand-matched) of a PolyASite 2.0 representative site; it is agreement with a curated atlas, not ground truth (atlas-novel true sites count as false positives). R_{det} denominator = atlas sites in genes detected in the dataset (285,136 human / 126,686 mouse); full-atlas recall of the default 0.089 (PBMC; 569,005) and 0.098 / 0.100 (mice; 301,006). One scorer (score_tool.py), same denominators for every tool; 3-seed gene-body-shuffled nulls. scUTRquant is catalog-based (hollow, not ranked). Pre-registration: the precision-first default and its $P@100 \geq 0.50$ gate were committed before the ORIGINAL run and are unchanged here (gate 01:19, original arms 02:59, v2 re-run 16:14, all 2026-08-21; 13 §1, 19); the ≥ 2 -molecule threshold was pre-registered before that run (12 CORRECTION). Gate: PASS on all three (0.706 / 0.745 / 0.757); no ≥ 1 -molecule output clears the original $P \geq 0.38$ floor. The v1 run (code 4efeb125, 15) gave 0.717 / 0.741 / 0.755; the ~ 1.1 pp PBMC move is the corrected minus-strand IP window (#96), reconstructed key-for-key. The IP filter buys +5.3 pp atlas-agreement precision for ~ 1.2 pp R_{det} (mouse 1); compute 12.5 GB / ~ 28 -35 min on PBMC (was 294 GB / 3:46 h); 19 §1/§3. Caveats: single PBMC donor / BAM; the two mice are one study and chemistry; the default's recall is at or below polyApipe's ($\sim 12\%$ PBMC, $\sim 18\%$ mouse) and the $F1_{det}$ lead (+0.020 / +0.014) is not a meaningful margin — the claim is precision, not overall accuracy. The non-IP PBMC ≥ 2 -molecule file (pas_tier1_ge2mol_nolP_POSTHOC) is not the default and is not shown. Verified: final_v2_verify/VERIFIED_v2.md (FIXED). Every plotted value: results/figures/manuscript/final_benchmark*.tsv.